🎬 RENSTO CONTENT AI - System Overview

Status: 🚧 In Development **Launch Date**: TBD (6-8 weeks) **Service Type**: Ready Solutions - Content Creator Package **Target Market**: YouTubers, Content Creators, Marketing Agencies

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Executive Summary

Rensto Content AI is an AI-powered content processing system that transforms any content (YouTube videos, PDFs, documents, podcasts) into viral scripts, summaries, blog posts, and provides RAG-powered chatbot capabilities.

Value Proposition: *"Transform any content into viral gold in minutes"*

Core Features

1. Multi-Format Content Ingestion

- YouTube video transcription (via Apify + Gemini)
- PDF/Document extraction (via Gemini multimodal)
- Image OCR processing
- Audio transcription
- Web page scraping (via FireCrawl)

2. RAG-Powered Content Chat

- Chat with your uploaded content
- Context-aware AI responses
- Multi-turn conversations with memory
- Source citations
- Powered by LightRAG + OpenAI

3. Viral Content Generation

- TikTok scripts (30-60 sec)
- Instagram Reel scripts (15-30 sec)
- YouTube Short scripts (60 sec)
- Twitter threads (10 tweets)
- Blog posts (1500-2000 words)
- Email newsletters

4. Content Library & Management

- Organized content dashboard
- Search across all content
- Tag and categorize
- Export options

Technical Architecture

Infrastructure Stack

```
┌──────────────────────────────────────┐
│      RENSTO CONTENT AI SYSTEM        │
├──────────────────────────────────────┤
│                                      │
│  ┌──────────────────────────────┐    │
│  │  Customer Portal (Next.js)   │    │
│  │  - Upload Interface          │    │
```

```
│  │  - Chat Interface              │   │
│  │  - Content Dashboard           │   │
│  │  - Generator Tools             │   │
│  └────────────────────────────────┘   │
│              ↓                         │
│  ┌────────────────────────────────┐   │
│  │  n8n Workflows                 │   │
│  │  - RDY-CONTENT-001: YouTube    │   │
│  │  - RDY-CONTENT-002: Documents  │   │
│  │  - RDY-CONTENT-003: RAG Chat   │   │
│  │  - RDY-CONTENT-004: Scripts    │   │
│  │  - RDY-CONTENT-005: Blogs      │   │
│  └────────────────────────────────┘   │
│              ↓                         │
│  ┌────────────────────────────────┐   │
│  │  AI Processing Layer           │   │
│  │  - OpenAI GPT-4o-mini (primary) │   │
│  │  - OpenAI text-embed-3-small   │   │
│  │  - Gemini API (transcription)  │   │
│  │  - LightRAG (vectors)          │   │
│  └────────────────────────────────┘   │
│              ↓                         │
│  ┌────────────────────────────────┐   │
│  │  Data Storage                  │   │
│  │  - n8n Data Tables (primary)   │   │
│  │  - Airtable (dashboards)       │   │
│  └────────────────────────────────┘   │
└────────────────────────────────────────┘
```

Technology Stack

AI/ML:

- **OpenAI** (cloud LLMs)
  - `gpt-4o-mini` - Content generation ($0.15/$0.60 per 1M tokens)
  - `text-embedding-3-small` - Embeddings ($0.02 per 1M tokens)
- **Gemini API** - Transcription, OCR, fallback processing
- **LightRAG** - Vector storage and retrieval

Processing:

- **n8n** - Workflow automation engine
- **Apify** - YouTube transcript extraction
- **FireCrawl** - Web scraping
- **Tavily** - Research API

Storage:

- **n8n Data Tables** - Primary operational data (Tier 1)
- **Airtable** - Customer dashboards and reporting (Tier 2)
- **Local Files** - LightRAG vector indices

Infrastructure:

- **VPS**: 173.254.201.134 (existing n8n server)
- **Docker**: n8n container
- **HTTPS**: Cloudflare Tunnel (n8n.rensto.com)
- **AI**: OpenAI API (no local hosting required)

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n8n Workflows

RDY-CONTENT-001: YouTube Video Processor

Purpose: Extract and process YouTube video transcripts

Flow:

```
Webhook (YouTube URL) →
HTTP Request (metadata) →
Apify (transcript) →
Code (clean) →
Text Splitter (1000 chars) →
OpenAI Embeddings →
LightRAG Insert →
Store in Data Table →
Respond
```

Input: `{ url: "https://youtube.com/watch?v=..." }` **Output**: `{ transcript, word_count, processing_time }`

RDY-CONTENT-002: PDF/Document Processor

Purpose: Extract text from documents

Flow:

```
Webhook (file upload) →
Read Binary File →
Gemini (extract text) →
Text Splitter →
OpenAI Embeddings →
LightRAG Insert →
Store in Data Table →
Respond
```

Supported Formats: PDF, DOCX, images (OCR), TXT

RDY-CONTENT-003: RAG Chat Handler

Purpose: Chat with uploaded content

Flow:

```
Webhook (chat message) →
LightRAG Query (retrieve context) →
AI Agent (OpenAI GPT-4o-mini) →
  - System: "Use context to answer"
  - Tools: [LightRAG Retriever]
  - Memory: Window Buffer (10 messages)
Store chat →
Respond
```

Features:

- Multi-turn conversations
- Context-aware responses

- Source citations
- Conversation history

RDY-CONTENT-004: Viral Script Generator

Purpose: Generate platform-specific scripts

Flow:

```
Webhook (content + platform) →
AI Chain (OpenAI GPT-4o-mini) →
  Prompt: "Generate {platform} script..."
Store generated script →
Respond
```

Platforms Supported:

- **TikTok**: 150 words, 30-60 sec, vertical format
- **Instagram Reel**: 100 words, 15-30 sec
- **YouTube Short**: 200 words, 60 sec
- **Twitter Thread**: 280 chars × 10 tweets

Script Structure:

1. Hook (first 3 seconds)
2. 3 key points
3. Call-to-action

RDY-CONTENT-005: Blog/Newsletter Generator

Purpose: Generate long-form content

Flow:

```
Webhook (topic + sources) →
Tavily (research) →
LightRAG Query (user content) →
AI Chain (OpenAI GPT-4o-mini) →
  Prompt: "Write SEO blog post..."
Store generated blog →
Respond
```

Blog Structure:

- Title & meta description
- Introduction with hook
- 5 main sections (H2/H3)
- Conclusion with CTA
- 1500-2000 words
- SEO optimized

Data Models

n8n Data Tables

`content_uploads`

```
{
  id: string (auto)
  customer_id: string
  type: 'youtube' | 'pdf' | 'document' | 'audio'
  url: string
  status: 'processing' | 'completed' | 'failed'
  file_size: number
  created_at: datetime
}
```

transcripts

```
{
  id: string (auto)
  upload_id: string (FK → content_uploads)
  content: text
  word_count: number
  language: string
  created_at: datetime
}
```

generated_content

```
{
  id: string (auto)
  upload_id: string (FK → content_uploads)
  type: 'script' | 'blog' | 'newsletter' | 'thread'
  platform: string
  content: text
  word_count: number
  created_at: datetime
}
```

chat_sessions

```
{
  id: string (auto)
  customer_id: string
  messages: json[]
  last_message_at: datetime
  created_at: datetime
}
```

rag_embeddings

```
{
  id: string (auto)
  content_id: string (FK → content_uploads)
  chunk_index: number
  chunk_text: text
  embedding: vector (stored in LightRAG)
  metadata: json
```

```
  created_at: datetime
}
```

Pricing Model

Three Tiers

Basic - $297/month

- 50 content uploads/month
- 10 GB storage
- YouTube + PDF processing
- RAG chat (1000 messages/month)
- 50 script generations/month
- Email support
- **Target**: Individual creators

Professional - $697/month ⭐ **RECOMMENDED**

- 250 content uploads/month
- 50 GB storage
- All content types
- RAG chat (unlimited messages)
- 250 script generations/month
- 50 blog posts/month
- Priority support
- Slack integration
- **Target**: Agencies, teams

Enterprise - $1,997/month

- Unlimited uploads
- 500 GB storage
- All features
- White-label option
- Custom workflows
- Priority OpenAI processing
- API access
- Dedicated support
- **Target**: Large agencies, brands

Add-ons

- **Extra Storage**: $10/10GB
- **Extra Processing**: $50/100 uploads
- **Custom Model Training**: $500 one-time
- **White-Label Setup**: $2,000 one-time

API Endpoints

Base URL

`https://api.rensto.com/v1/content-ai`

Authentication

```
Authorization: Bearer {customer_token}
```

Endpoints

`POST /upload`

Upload content for processing

Request:

```
{
  "type": "youtube",
  "url": "https://youtube.com/watch?v=..."
}
```

Response:

```
{
  "upload_id": "uuid",
  "status": "processing",
  "estimated_time": "2 minutes"
}
```

`POST /process/youtube`

Process YouTube video

Request:

```
{
  "url": "https://youtube.com/watch?v=...",
  "options": {
    "extract_chapters": true,
    "generate_summary": true
  }
}
```

`POST /process/document`

Process document (PDF, DOCX, image)

Request:

```
{
  "file": "base64_encoded_file",
  "type": "pdf"
}
```

`POST /chat`

Chat with RAG-powered bot

Request:

```
{
  "session_id": "uuid",
  "message": "Summarize this video",
```

```
  "context": ["upload_id_1", "upload_id_2"]
}
```

Response:

```
{
  "response": "Based on the video, ...",
  "sources": ["video_1_chunk_3", "video_1_chunk_7"],
  "confidence": 0.89
}
```

`POST /generate/script`

Generate viral script

Request:

```
{
  "content_id": "upload_id",
  "platform": "tiktok",
  "tone": "enthusiastic",
  "length": "30_seconds"
}
```

Response:

```
{
  "script": "🎬 Hook: Did you know...\n\n📌 Point 1: ...\n\n...",
  "word_count": 150,
  "estimated_duration": "32 seconds"
}
```

`POST /generate/blog`

Generate blog post

Request:

```
{
  "topic": "Content Marketing Trends 2025",
  "sources": ["upload_id_1", "upload_id_2"],
  "keywords": ["SEO", "video marketing"],
  "tone": "professional"
}
```

`GET /library/:customerId`

Get customer's content library

Response:

```
{
  "uploads": [...],
  "total_count": 247,
```

```
  "storage_used": "12.4 GB",
  "monthly_limit": "50 GB"
}
```

Webhook URLs

Production (n8n.rensto.com)

- **YouTube**: `https://n8n.rensto.com/webhook/content-youtube`
- **Document**: `https://n8n.rensto.com/webhook/content-document`
- **Chat**: `https://n8n.rensto.com/webhook/content-chat`
- **Script Gen**: `https://n8n.rensto.com/webhook/content-script`
- **Blog Gen**: `https://n8n.rensto.com/webhook/content-blog`

Performance Metrics

Target SLAs

Processing Times:

- YouTube video (10 min): <2 minutes
- PDF (50 pages): <1 minute
- RAG chat response: <3 seconds
- Script generation: <10 seconds
- Blog generation: <30 seconds

Availability:

- Uptime: 99.5%
- API response time: <500ms (p95)

Storage:

- Per customer average: 20 GB
- Text storage (n8n): Unlimited
- Vector storage (LightRAG): 100GB total

Business Metrics

Year 1 Goals

Customers:

- Month 3: 10 customers (beta)
- Month 6: 50 customers
- Month 12: 150 customers

Revenue (assuming avg $697/customer):

- Month 3: $6,970 MRR
- Month 6: $34,850 MRR
- Month 12: $104,550 MRR = **$1.25M ARR**

Margins:

- Infrastructure costs: $50-500/month (OpenAI API, scales with usage)
- Gross margin: 70-85% (varies with volume)
- Net profit margin: 55-70% (after support and API costs)

Implementation Status

Phase 1: Infrastructure Setup (Week 1) ✅

- [x] Evaluate AI providers (Ollama vs OpenAI)
- [x] Decided on OpenAI API for speed-to-market
- [x] OpenAI credentials configured in n8n
- [ ] Create n8n Data Tables
- [ ] Test basic AI workflow with OpenAI

Phase 2: Core Workflows (Weeks 2-3)

- [ ] Build RDY-CONTENT-001 (YouTube)
- [ ] Build RDY-CONTENT-002 (Documents)
- [ ] Build RDY-CONTENT-003 (RAG Chat)
- [ ] Build RDY-CONTENT-004 (Scripts)
- [ ] Build RDY-CONTENT-005 (Blogs)

Phase 3: Customer Portal (Week 4)

- [ ] Build upload interface
- [ ] Build chat interface
- [ ] Build content dashboard
- [ ] Build generator tools
- [ ] API integration

Phase 4: Documentation & Launch (Weeks 5-6)

- [x] Technical documentation (this file)
- [ ] User guide
- [ ] API documentation
- [ ] Marketing materials
- [ ] Beta launch (10 customers)

Competition Analysis

Poppy AI

Pricing: $300-399/year ($25-27/month) **Strengths**: Visual canvas interface, multi-AI models **Weaknesses**: Manual process, no automation

Rensto's Advantage:

- **Lower price**: $297/month vs $399/year (but more features)
- **Automation**: Workflows automate entire process
- **RAG included**: Poppy doesn't have RAG chat
- **Brand**: Rensto's automation expertise

Other Competitors

- **Descript**: $30/month (focused on video editing)
- **Jasper**: $49/month (focused on copywriting)
- **ChatGPT Plus**: $20/month (no specialized content tools)

Differentiation: We're the only solution combining **content processing + automation + RAG + generation** in one platform.

Security & Compliance

Data Protection

- All content encrypted at rest
- HTTPS for all API calls
- Per-customer data isolation
- GDPR compliant data handling

Content Moderation

- Terms of service enforcement
- Copyright violation checks
- Spam/abuse detection
- Content filtering (optional)

Support & Training

Customer Onboarding

1. **Welcome Email** (automated)
2. **Setup Call** (15 min)
3. **First Upload Walkthrough**
4. **Weekly Check-in** (first month)

Documentation

- Video tutorials (YouTube)
- Written guides (Notion)
- API documentation (this file)
- FAQ & troubleshooting

Support Channels

- **Basic**: Email (24hr response)
- **Pro**: Email + Slack (12hr response)
- **Enterprise**: Email + Slack + Phone (4hr response)

Future Roadmap

Q2 2025 (Post-Launch)

- Audio podcast processing
- Collaboration features (teams)
- Advanced analytics dashboard
- A/B testing for content

Q3 2025

- Multi-language support
- Mobile app (iOS/Android)
- Video generation (AI-powered)
- Integration marketplace

Q4 2025

- White-label platform
- Agency partnerships
- API marketplace
- Education/courses

Contact & Resources

Project Lead: Shai Friedman (service@rensto.com) **Technical Documentation**: `/docs/products/CONTENT_AI_TECHNICAL_DOCUMENTATION.md` **User Guide**: `/docs/products/CONTENT_AI_USER_GUIDE.md` **API Docs**: This file (API Endpoints section)

n8n Instance: https://n8n.rensto.com **Customer Portal**: https://rensto.com/content-ai (coming soon)

Last Updated: 2025-10-03 **Version**: 1.0.0-alpha **Status**: 🚧 In Development